

Project Demonstration: Rubik's Cube Solving Machine

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Project Specifications:

Constraints

- ☐ Meeting safety standards
- ☐ A substantial mechanical re-design of the EV3 robot in standard configuration
- ☐ Uses of 3 motors (all of which have controlled movement)
- ☐ Uses of 4 distinct types of inputs:
 - Standard sensors:
 - Ultrasonic sensor
 - colour sensor
 - motor encoders
 - EV3 buttons are used to control system processes
 - Pressing the **Left** or **Right** button to scramble the cube
 - Pressing the **Enter** button to colour sense the cube and then followed by another **Enter** button to solve the cube
 - Pressing the **Down** button after one task is completed will close the system
 - RobotC File I/O
- ☐ Use of timers
 - Time the robot taken to solve the cube is displayed on the console at the end
- ☐ Software uses repetition (loops)
- ☐ Software uses decisions (if statements)
- ☐ Software contains six non-trivial functions, not including main,
 - at least one function return something
 - at least one function having parameters

Criteria

- ☐ Robot accepts a 3X3 cube in any pattern and can solve the cube
- ☐ Robot can scramble the cube to any random pattern
- ☐ Cube is rotated/worked stably and lands in correct position
- ☐ Cube is solved as efficiently as possible given the constraints

Demo Tasks:

- ☐ Robot is set to the initial position manually (meaning that the robot arm is at rest, the base is at 90 degrees and the colour sensor is at the very back)
- ☐ After starting the program, the user is able to choose from pressing the Left or Right buttons to scramble the cube or pressing the Enter button to colour sense then solve. If no button is pressed, nothing happens.
- ☐ Ultrasonic sensor must sense an existence of a cube to start any actions. No action is performed if the cube is not in place even after buttons are pressed.
- ☐ After the cube is placed, and either Left or Right button is pressed, the robot will start scrambling the cube to a randomized pattern.
- ☐ Now, by pressing the Enter button, the robot will colour sense all faces of the cube
- ☐ Using the algorithm in Python and reading the current state of the cube, a solution to solve the cube will be provided shortly. Robot will read this solution to solve the cube. (Algorithm is not part of the grading process, however the Robot C File I/O will be)
- ☐ By pressing the Enter button again, the robot should begin to solve the cube. Robot will complete a series of moves (flips, rotating the base and layers) until the cube is fully solved.
- ☐ Time the robot takes to solve the cube is displayed on the console
- ☐ User can press the Down button to end the program.

Project Work Declaration

We declare that the design, including configuration of the robot and the software, in this project is our original work completed during the current school term for the MTE 100/MTE 121 course project except as noted below:

[1] M. Beke, "Online Rubik's Cube Solver," *Rubik's Cube Solver*. <https://rubiks-cube-solver.com/>.

[2] FarhanShoukat, "GitHub - FarhanShoukat/Rubiks-Cube-Solver: a Comprehensive Comparision of IDFS and IDA* in Solving Rubik's Cube," *GitHub*, 2018.
<https://github.com/FarhanShoukat/Rubiks-Cube-Solver>

[3] Selenium, "Selenium WebDriver," [Online]. Available: <https://www.selenium.dev>. Accessed: Nov. 19, 2024].

[4] David Gilday, "MindCub3r for EV3," *Mindcuber.com*, 2020.
<https://www.mindcuber.com/mindcub3r/mindcub3r.html>.

We declare that we have neither given nor received an electronic copy or a printed version of any part of this design except as stated above.

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